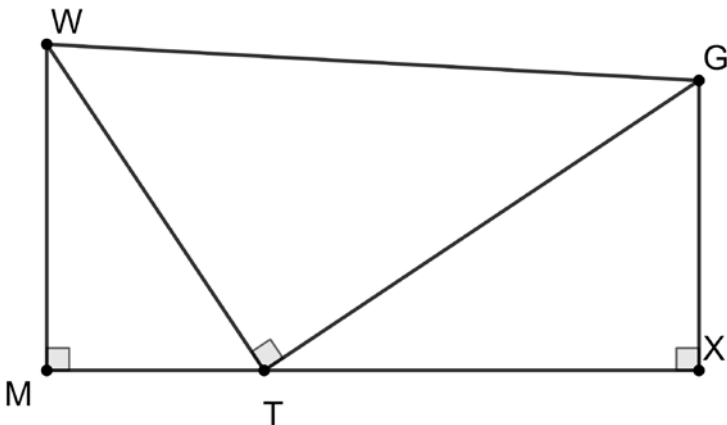


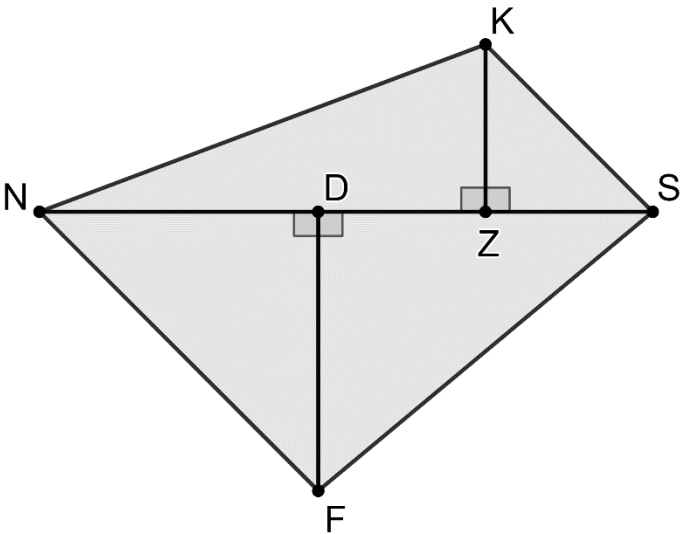
Right triangles WMT , GTW , and TXG are shown where $TX = 8$, $m\angle XTG = 34^\circ$, and $m\angle TGW = 37^\circ$.



Complete the table. In the work column, write the equation or trigonometric ratio used to determine the side measure. If necessary, round to the nearest thousandth.

	Side	Work	Side Measure
1.	$\overline{XG}$	$\tan 34 = \frac{XG}{8}$	5.396
2.	$\overline{TG}$	$\cos 34 = \frac{8}{TG}$	9.650
3.	$\overline{WT}$	$\tan 37 = \frac{WT}{9.650}$	7.272
4.	$\overline{WM}$	$\sin 56 = \frac{WM}{7.272}$	6.028
5.	$\overline{MT}$	$\cos 56 = \frac{MT}{7.272}$	4.066

Quadrilateral $NKSF$ is composed of right triangles NDF , FDS , SZK , and NZK . $ZK = 3$, $DF = 5$, $m\angle DNF = 45^\circ$, $m\angle SKZ = 45^\circ$, $m\angle FSD = 40^\circ$, and $m\angle ZNK = 21^\circ$.



Complete the table. In the work column, write the equation or trig ratio used to determine the side measure. If necessary, round to the nearest thousandth.

	Side	Work	Side Measure
6.	$\overline{KS}$	$\cos 45 = \frac{3}{KS}$	4.243
7.	$\overline{FS}$	$\sin 40 = \frac{5}{FS}$	7.779
8.	$\overline{NF}$	$\sin 45 = \frac{5}{NF}$	7.071
9.	$\overline{NK}$	$\sin 21 = \frac{3}{NK}$	8.371

10. What is the perimeter, to the nearest thousandth, of quadrilateral $NKSF$?
 $4.243 + 7.779 + 7.071 + 8.371 = 27.464$